

BUKHARA KOGON, Uzbekistan

WMO: 386830

Lat: 39.717N

Lon: 64.617E

Elev: 224

StdP: 98.66

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-7.9	-15.9	1.0	-9.4	-12.6	1.3	-5.0	9.1	1.9	7.7	4.7	3.4	0	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	39.9	20.0	38.5	19.6	37.2	19.2	21.9	35.9	21.0	35.3	20.2	34.4	5.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.5	28.2	15.8	11.5	27.2	14.7	10.7	27.0	65.1	36.1	61.9	35.2	59.2	34.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.1	7.8	7.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-12.3	42.8	4.0	1.7	-15.2	44.0	-17.6	45.0	-19.8	45.9	-22.7	47.1
			-12.7	23.3	4.1	1.4	-15.6	24.3	-18.0	25.1	-20.3	25.9	-23.2	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	2.2	4.3	10.7	17.3	23.4	28.1	29.6	27.9	22.1	15.3	8.3	3.2		
	DBStd	10.64	5.30	5.58	4.98	4.79	3.39	2.72	2.39	2.80	3.44	4.37	5.01	4.64		
	HDD10.0	781	245	168	52	5	0	0	0	0	0	9	88	214		
	HDD18.3	2101	500	393	239	74	5	0	0	0	8	111	303	468		
	CDD10.0	3007	3	9	75	223	414	542	608	555	363	174	37	4		
	CDD18.3	1287	0	0	3	43	161	292	350	297	121	18	2	0		
	CDH23.3	17158	0	0	56	510	1966	4140	5011	3832	1366	255	23	0		
	CDH26.7	9683	0	0	10	183	953	2497	3108	2252	610	66	4	0		
(14) Wind	WSAvg	3.6	3.2	3.6	3.7	3.8	3.8	4.1	4.5	4.3	3.9	3.1	2.8	3.1		
(15) Precipitation	PrecAvg	139	18	21	26	22	12	2	0	0	1	2	14	16		
	PrecMax	212	42	48	50	55	59	11	5	2	8	6	43	36		
	PrecMin	68	4	7	5	1	0	0	0	0	0	0	0	2		
	PrecStd	42	11	11	14	15	15	3	1	0	2	2	11	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	21.0	28.3	34.5	37.6	40.9	42.4	40.7	36.7	31.6	26.5	18.7		
		MCWB	9.6	11.6	14.4	17.2	18.4	19.5	21.0	20.8	18.7	16.4	14.1	9.6		
	2%	DB	14.5	17.5	24.8	31.2	35.5	39.2	40.3	38.8	34.2	28.4	21.7	15.4		
		MCWB	8.3	10.2	13.5	16.4	17.7	19.2	20.4	20.3	17.5	14.8	12.4	8.6		
	5%	DB	11.8	14.8	21.9	28.5	33.7	37.8	38.8	37.4	32.4	26.1	18.8	12.6		
		MCWB	7.2	8.8	12.5	15.6	17.1	18.8	20.0	19.6	16.9	14.3	11.3	7.5		
	10%	DB	9.2	12.4	19.2	26.3	31.8	36.3	37.3	35.8	30.3	24.0	16.4	10.1		
		MCWB	5.9	8.0	11.5	14.8	16.7	18.4	19.5	18.9	16.2	13.4	10.0	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	12.6	15.8	19.7	20.4	22.4	23.5	22.9	20.2	17.5	14.9	10.9		
		MCDB	15.3	18.3	23.6	29.2	31.6	35.1	37.7	37.2	33.9	28.6	23.3	16.0		
	2%	WB	8.7	11.0	14.5	17.9	19.3	21.0	22.1	21.6	18.6	15.9	13.3	9.2		
		MCDB	13.5	16.2	22.1	26.8	30.6	35.3	36.5	35.9	31.6	26.1	20.3	14.2		
	5%	WB	7.6	9.5	13.2	16.7	18.5	20.1	21.1	20.7	17.6	14.8	11.8	7.9		
		MCDB	11.2	14.0	20.5	25.9	29.9	34.4	35.8	34.8	30.4	24.8	17.6	12.0		
	10%	WB	6.2	8.1	12.0	15.6	17.7	19.4	20.4	19.8	16.7	13.8	10.5	6.6		
		MCDB	9.2	11.9	18.5	24.0	29.2	33.3	35.0	33.8	29.1	22.9	15.5	10.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.6	11.2	12.6	14.1	14.9	14.5	14.8	14.8	14.0	11.3	8.9		
		MCDBR	12.8	13.9	15.5	16.3	16.8	16.9	16.3	16.2	16.8	16.8	14.8	13.2		
		MCWBR	7.7	7.5	6.5	5.7	5.4	5.6	5.8	6.3	6.4	7.3	7.4	7.8		
	5% WB	MCDBR	11.4	12.0	13.8	14.0	14.5	15.2	15.2	15.5	16.1	15.3	12.8	11.9		
		MCWBR	7.3	7.1	6.3	5.5	5.3	5.5	5.7	6.4	6.7	7.1	6.7	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.381	0.422	0.492	0.520	0.516	0.509	0.474	0.444	0.425	0.466	0.441	0.383		
	taud		2.121	2.019	1.873	1.839	1.873	1.892	1.987	2.053	2.062	1.923	1.975	2.098		
	Ebn at Noon		746	768	753	763	778	783	807	821	809	707	661	704		
	Edh at Noon		111	142	183	201	199	195	176	160	149	154	128	106		
(44) All-Sky Solar Radiation	RadAvg		1.97	2.90	4.11	5.56	6.86	7.77	7.69	7.01	5.72	3.95	2.48	1.75		
	RadStd		0.18	0.29	0.32	0.46	0.30	0.23	0.16	0.14	0.15	0.21	0.23	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	-0.87	-1.29	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.32	N/A	-0.80	N/A	N/A	N/A

Nomenclature: See separate page